

Practice set 3

Quadratics and the discriminant

Write in the spaces

Answers · official assessment schedule extracts on the last pages

- (b) A quadratic equation has solutions of $x = -\frac{2}{3}$ and $x = 4$.

Find the original equation, giving your answer in the form of $ax^2 + bx + c = 0$, where a , b , and c are whole numbers.

- (c) Consider a quadratic equation in the form $x^2 - 3kx + 2k^2 = 0$, where k is a non-zero constant.

Show that one solution is twice the other solution.

QUESTION TWO

- (a) A quadratic equation, $ax^2 + bx + c = 0$, has solutions of $\frac{1}{3}$ and $\frac{-2}{7}$.

Find the values of the integers a, b, and c.

a = _____ b = _____ c = _____

- (b) (i) What is the discriminant of the equation $2x^2 - 12x + 7 = 0$?

- (ii) Suppose $y = 2x^2 - 12x + k$, where k is a constant.

For what value of k will the equation $y = 0$ have exactly one solution?

Official assessment schedule extract · pages 1 and 2

Q ONE	Expected coverage	Achievement (u)	Merit (r)	Excellence (t)
(b)	Deduce that $(3x + 2)(x - 4) = 0$ OR $\left(x + \frac{2}{3}\right)(x - 4) = 0$ so Original equation was $3x^2 - 10x - 8 = 0$	Correct equation but coefficients not whole numbers OR incorrect relevant factorised form but consistent expanded form.	Correct equation.	
(c)	$x^2 - 3kx + 2k^2 = 0$ $(x - k)(x - 2k) = 0$ So solutions are k and $2k$, [hence one is twice the other]. OR $x^2 - 3kx + 2k^2 = 0$ $\Delta = (-3k)^2 - 4(1)(2k^2) = k^2$ Hence, $x = \frac{-(-3k) \pm \sqrt{k^2}}{2}$ $= \frac{3k \pm k}{2}$ $= \frac{4k}{2}$ or $\frac{2k}{2}$ i.e. one root is $2k$ and the other is k and hence one is twice the other.	Equation factorised. OR Discriminant evaluated.	Correct conclusion with valid working.	

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Q	Evidence	Achievement	Merit	Excellence
TWO (a)	$\left(x - \frac{1}{3}\right)\left(x + \frac{2}{7}\right)$ $= (3x - 1)(7x + 2)$ $= 21x^2 - x - 2$ $a = 21, b = -1, c = -2$	Correct values of a, b, and c.		
(b)(i)	$(-12)^2 - 4(2)(7) = 88$	Correct discriminant OR		
(ii)	So $(-12)^2 - 4(2)(k) = 0$ $8k = 144$ $k = 18$ accept use of inequality	substitution made (line 1)	Correct value of k.	